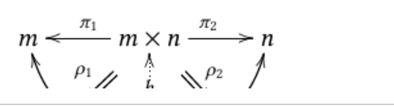
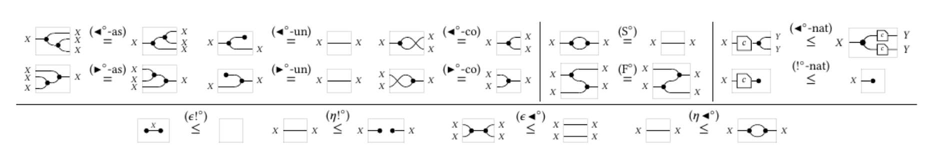
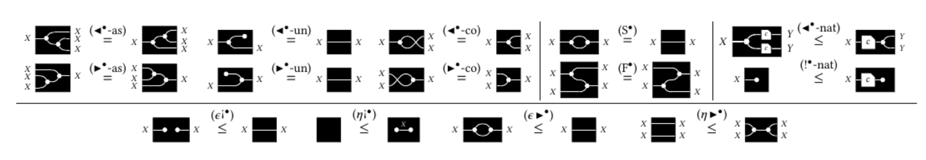
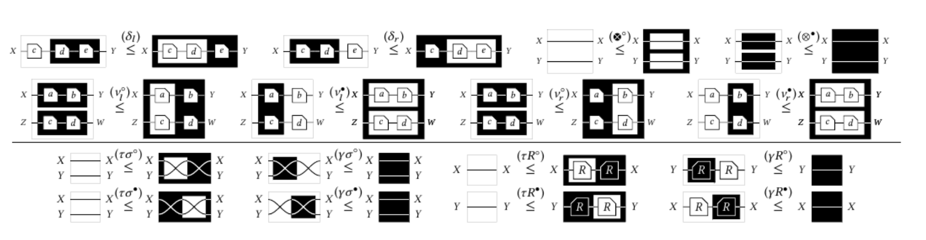
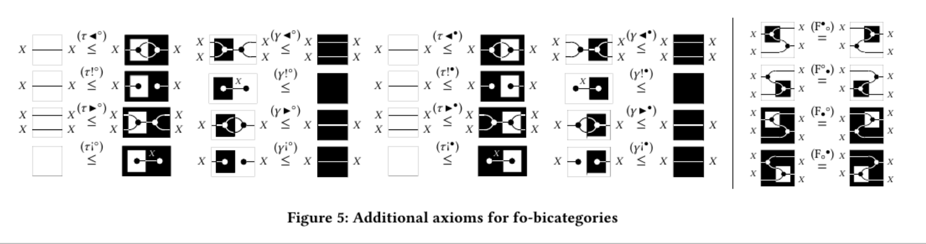
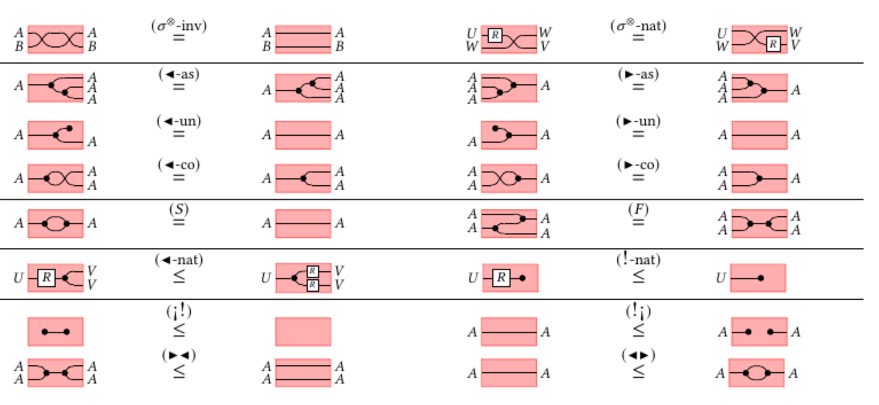
Axiom crosscheck — source figure against Lean field

generated by `scripts/paper-figs`; papers cited by file name, never by arXiv id

Left column: the figure as printed, clipped from the paper at 150 dpi. Right column: the docstring of the Lean declaration transcribed from it, verbatim. Reading a row across is the check — the picture and the equation number in the docstring must be the same equation.

Cartesian bicategories of relations

source clip	Lean declaration	its docstring
	`Freyd.Diag.CartBicat.Δ` functorialSemanticsForRelationalTheories.pdf p. 17	no docstring — declaration not found in `diag/`
	`Freyd.Diag.CartBicat.«!»` functorialSemanticsForRelationalTheories.pdf p. 17	no docstring — declaration not found in `diag/`
	`Freyd.Diag.CartBicat.«∇»` functorialSemanticsForRelationalTheories.pdf p. 17	no docstring — declaration not found in `diag/`
	`Freyd.Diag.CartBicat.«?»` functorialSemanticsForRelationalTheories.pdf p. 17	no docstring — declaration not found in `diag/`
r commutative monoids ca an additional commutative es, the models in $\mathbf{Rel}_\times$ are or is are monoid homomorphi	`Freyd.Diag.CartBicat.«∇_assoc»` functorialSemanticsForRelationalTheories.pdf p. 8	Associativity of `∇`; Example 2.3(a) eq. (5).
e thought of as the prop of ck" monoid. Again, as in the	`Freyd.Diag.CartBicat.«∇_comm»` functorialSemanticsForRelationalTheories.pdf p. 8	Commutativity of `∇`; eq. (6).
ialgebras, ase of lax	`Freyd.Diag.CartBicat.«∇_unit»` functorialSemanticsForRelationalTheories.pdf p. 8	Unit law for `(∇, ?)`; eq. (7).
ius theories give us much ries in $\mathbf{Rel}_\times$: they are much into the fold. This paper intr oles.	`Freyd.Diag.CartBicat.Δ_assoc` functorialSemanticsForRelationalTheories.pdf p. 8	no docstring — declaration not found in `diag/`
that a way of doing "resour expressive and allow us to l es the basic theory together	`Freyd.Diag.CartBicat.Δ_comm` functorialSemanticsForRelationalTheories.pdf p. 8	Cocommutativity of `Δ`; eq. (9).
e-honest ing many	`Freyd.Diag.CartBicat.Δ_counit` functorialSemanticsForRelationalTheories.pdf p. 8	Counit law for `(Δ, !)`; eq. (10).
important tool in categorical models of compu inistic from the probabilistic models, and class	`Freyd.Diag.CartBicat.frob_left` functorialSemanticsForRelationalTheories.pdf p. 8	(41), left form: `(𝟙 ⊗ Δ);α⁻¹;(∇ ⊗ 𝟙) = ∇;Δ`.
whether to separate quantum [20, 43],	`Freyd.Diag.CartBicat.special` functorialSemanticsForRelationalTheories.pdf p. 8	The SPECIAL law `Δ;∇ = 𝟙` (functorialSemanticsForRelationalTheories.pdf p. 18, eq. (12)). Not an axiom: `𝟙 ≤ Δ;∇` is (38), and the reverse is the paper's displayed derivation — weaken the identity on one strand of the bubble to `!;?` by (40), then collapse with the counit and unit laws.
and n, a *lax product* is an object $m \times n$ $\pi_2 : m \times n \longrightarrow n$ such that, for any f : nd 2-cells ρ_1, ρ_2 satisfying	`Freyd.Diag.CartBicat.«∇Δ_≤_𝟙»` functorialSemanticsForRelationalTheories.pdf p. 18	(37): `∇;Δ ≤ 𝟙`. With (38), `Δ ⊣ ∇`.
h *projections*, that is $\to m$, $g : k \longrightarrow n$, there	`Freyd.Diag.CartBicat.«𝟙_≤_Δ∇»` functorialSemanticsForRelationalTheories.pdf p. 18	(38): `𝟙 ≤ Δ;∇`. With (37), `Δ ⊣ ∇`.
	`Freyd.Diag.CartBicat.«?!_≤_𝟙»` functorialSemanticsForRelationalTheories.pdf p. 18	(39): `?;! ≤ 𝟙_I`. With (40), `! ⊣ ?`.
(32)	`Freyd.Diag.CartBicat.«𝟙_≤_!?»` functorialSemanticsForRelationalTheories.pdf p. 18	(40): `𝟙 ≤ !;?`. With (39), `! ⊣ ?`.
σ_2 which verify	`Freyd.Diag.CartBicat.frob_left` functorialSemanticsForRelationalTheories.pdf p. 18	(41), left form: `(𝟙 ⊗ Δ);α⁻¹;(∇ ⊗ 𝟙) = ∇;Δ`.
	`Freyd.Diag.CartBicat.frob_right` functorialSemanticsForRelationalTheories.pdf p. 18	(41), right form: `(Δ ⊗ 𝟙);α;(𝟙 ⊗ ∇) = ∇;Δ`.
unique $\xi : h' \Rightarrow h$ such that	`Freyd.Diag.CartBicat.lax_Δ` functorialSemanticsForRelationalTheories.pdf p. 18	(42): every arrow is lax for the comultiplication — `R;Δ ≤ Δ;(R ⊗ R)`. This is eq. (3), p. 4; Freyd has it product-free as §2.136's `R(S ∩ T) ⊑ RS ∩ RT` at `S := π₁°, T := π₂°`.
	`Freyd.Diag.CartBicat.lax_!` functorialSemanticsForRelationalTheories.pdf p. 18	(43): every arrow is lax for the counit — `R;! ≤ !`. Freyd: §2.152's step "`p_α` is maximal in `(α,λ)`, hence `R p_β ⊑ p_α`".
Indeed, considering the monoidal category c oduct as monoidal product) as codomain, th prop of commutative monoids are in bijectiv ive monoids. Here it is the products of **Set** th	`Freyd.Diag.CartBicat.lax_Δ` functorialSemanticsForRelationalTheories.pdf p. 4	(42): every arrow is lax for the comultiplication — `R;Δ ≤ Δ;(R ⊗ R)`. This is eq. (3), p. 4; Freyd has it product-free as §2.136's `R(S ∩ T) ⊑ RS ∩ RT` at `S := π₁°, T := π₂°`.
of sets and functions **Set** he symmetric monoidal ve correspondence with at ensure that, although	`Freyd.Diag.CartBicat.lax_!` functorialSemanticsForRelationalTheories.pdf p. 4	(43): every arrow is lax for the counit — `R;! ≤ !`. Freyd: §2.152's step "`p_α` is maximal in `(α,λ)`, hence `R p_β ⊑ p_α`".
d Walters). *Suppose that* **C** *has a lax product structure. Then*	`Freyd.Diag.CartBicat.special` functorialSemanticsForRelationalTheories.pdf p. 18	The SPECIAL law `Δ;∇ = 𝟙` (functorialSemanticsForRelationalTheories.pdf p. 18, eq. (12)). Not an axiom: `𝟙 ≤ Δ;∇` is (38), and the reverse is the paper's displayed derivation — weaken the identity on one strand of the bubble to `!;?` by (40), then collapse with the counit and unit laws.
 **Figure 2: Axioms of cartesian bicategories**	`Freyd.Diag.CartBicat` DiagrammaticAlgebraOfFirstOrderLogic.pdf p. 6	A CARTESIAN BICATEGORY OF RELATIONS (functorialSemanticsForRelationalTheories.pdf Def. 4.1).
 **Figure 3: Axioms of cocartesian bicategories**	`Freyd.Diag.CocartBicat` DiagrammaticAlgebraOfFirstOrderLogic.pdf p. 6	A COCARTESIAN BICATEGORY (`DiagrammaticAlgebraOfFirstOrderLogic.pdf` Fig. 3): `CartBicat` with the colours and the order inverted. Read every field against the `CartBicat` field of the same name in `diag/CB.lean`.
 **Figure 4: Axioms of closed symmetric monoidal linear bicategories**	`Freyd.Diag.LinearBicat` DiagrammaticAlgebraOfFirstOrderLogic.pdf p. 7	A LINEAR BICATEGORY (`DiagrammaticAlgebraOfFirstOrderLogic.pdf` Def. 5.1): two poset enriched categories with THE SAME objects, arrows and ordering but different identities and compositions, the white `(⨾°, id°)` — which is `Cat`'s own `≫` and `𝟙` — and the black `(⨾•, id•)`, such that `⨾°` linearly distributes over `⨾•`. Note what is NOT here: `id•` is not required to be `𝟙`, and `⨾•` is not required to agree with `≫` anywhere. In `Rel(Set)` they are as different as `∃` and `∀`.
 **Figure 5: Additional axioms for fo-bicategories**	`Freyd.Diag.FOBicat` DiagrammaticAlgebraOfFirstOrderLogic.pdf p. 9	A FIRST ORDER BICATEGORY (`DiagrammaticAlgebraOfFirstOrderLogic.pdf` Def. 6.1).
 Fig. 3. Axioms of cartesian bicategories	`Freyd.Diag.CartBicat` TapeDiagrams.pdf p. 6	A CARTESIAN BICATEGORY OF RELATIONS (functorialSemanticsForRelationalTheories.pdf Def. 4.1).

Sources recorded for later phases

No Lean counterpart yet: the tape layer (`diag/Tape.lean`) and the first-order layer (`diag/FO.lean`) are unwritten, and Example 2.3(d)(e) is the road Definition 4.1 does not take — the same monoid and comonoid combined as a bialgebra rather than as a Frobenius algebra.

cience. The use of monoidal structu
to be convenient for teaching and
ving new insights on foundational
time making the subject suitable fo

functorialSemanticsForRelationalTheories.pdf p. 8

ns [9, 10, 26, 8, 5, 31, 6]
in theory and practice
to a monograph [47],
ults of computability
n undergraduate level

functorialSemanticsForRelationalTheories.pdf p. 8

der this paper a humble offering of gratitude for the inspiration provided
and its academic environment.

functorialSemanticsForRelationalTheories.pdf p. 9

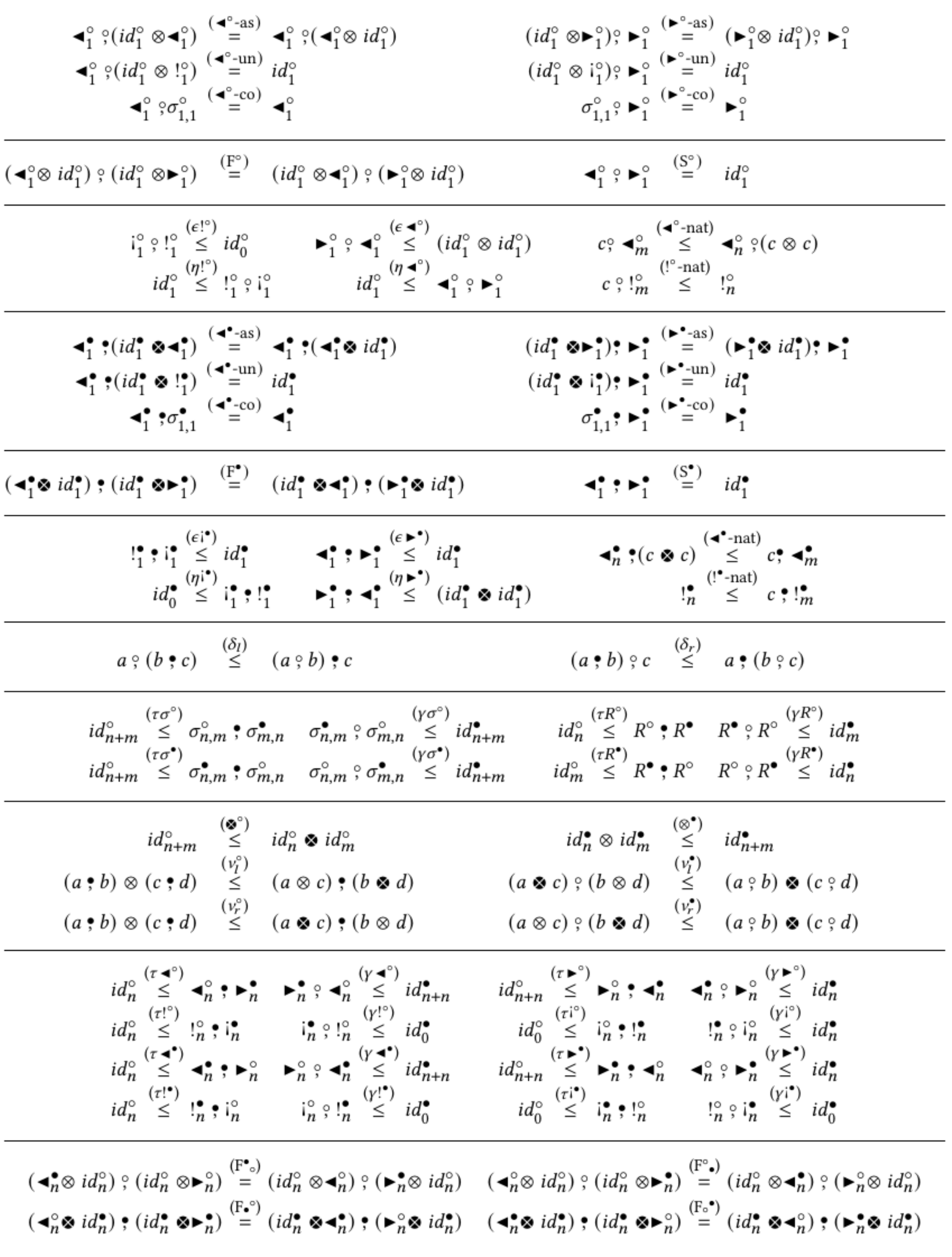


Figure 9: Axioms for NPR_Σ. Here a, b, c, d are diagrams of the appropriate type.

DiagrammaticAlgebraOfFirstOrderLogic.pdf p. 15

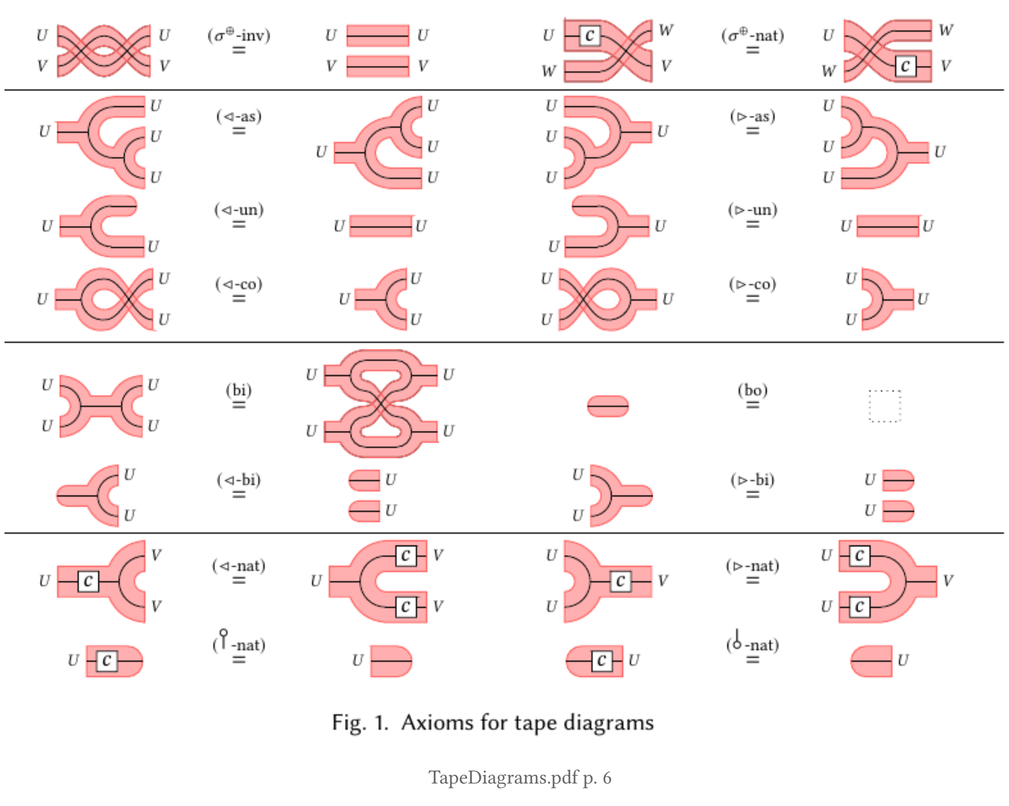


Fig. 1. Axioms for tape diagrams

TapeDiagrams.pdf p. 6

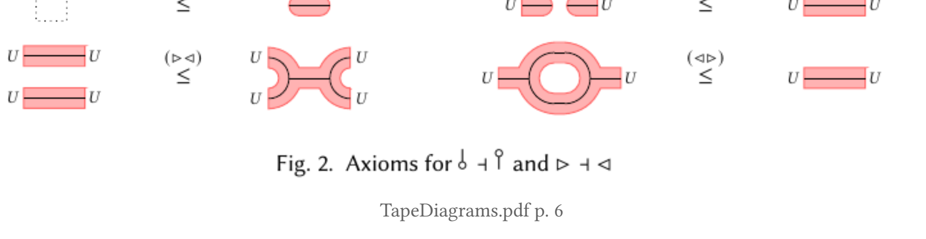


Fig. 2. Axioms for ⫰ ⊣ ⫯ and ⊳ ⊣ ⊲

TapeDiagrams.pdf p. 6